Unit

8 Covalent bonds

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Covalent bonds and molecules
- Chemical formulae of covalent compounds
- Predicting the formation of ionic and covalent compounds
- Dative covalent bonding
- Relative molecular mass and formula mass

Reference Website
Refer to a video clip on covalent bonding:
http://www.bbc.co.uk/schools/gcsebitesize/science/add_ocr_gateway/periodic_table/covalentbondingact.shtml
(accessed July 2014)

Animation
Covalent bond

8.1 Covalent bonds

In Unit 7, we have seen that sugar, distilled water and ethanol are non-conductors of electricity. All these compounds are made up of non-metals only and behave very differently from ionic compounds. The different behaviour is due to the structures that result when non-metallic atoms are held together by **covalent bonds**.

Covalent bonding involves electron sharing. Non-metallic atoms try to obtain the stable electronic arrangements of atoms of noble gases by sharing electrons. This results in the formation of a **molecule**.

The hydrogen molecule

A hydrogen atom has an electronic arrangement of 1. It needs one more electron to obtain the electronic arrangement of a helium atom (2). Each hydrogen atom can obtain the electronic arrangement of a helium atom by sharing its electron with another hydrogen atom. The pair of bonded hydrogen atoms is a hydrogen molecule (Fig. 8.1).

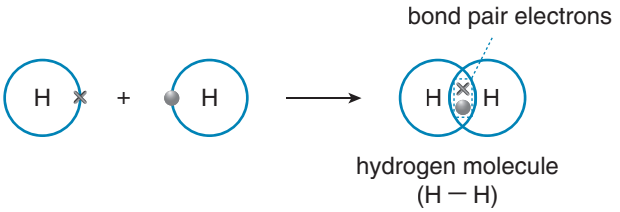


Fig. 8.1 Formation of covalent bond in a hydrogen molecule

The negatively charged electrons are attracted to the positively charged nuclei of both atoms. The resulting attraction is the covalent bond that holds the two atoms together (Fig. 8.2). Bonds in which only a pair of electrons is shared are called **single bonds**.

N1 Refer to page T36.

covalent bond 共價鍵　　molecule 分子　　single bond 單鍵

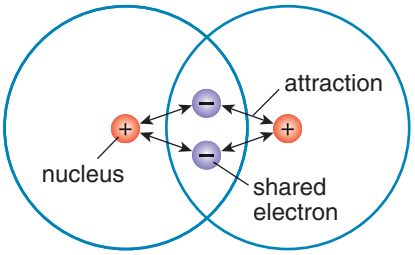


Fig. 8.2 A covalent bond is directional, acting solely between the two atoms involved in the bond

✓ A covalent bond is formed when one or more pairs of outermost shell electrons are shared between two atoms.

✓ A covalent bond is the strong electrostatic forces of attraction between the shared electrons and the two positively charged nuclei of the bonded atoms.

✓ The pair of electrons shared between two atoms is called **bond pair electrons**.

We can represent the covalent bond between atoms by a line drawn between the atoms. Hence a hydrogen molecule is represented as H–H.

Each hydrogen molecule consists of two hydrogen atoms. We can say hydrogen forms **diatomic molecules**. The chemical formula of hydrogen gas is H_2.

A molecule which consists of one atom only is called a **monoatomic molecule**. As noble gases exist as single atoms, we can also say noble gases exist as monoatomic molecules.

The chemical formula of a substance made up of molecules is also called its **molecular formula**. The molecular formula of a substance gives the actual number of atoms of each element in each molecule of the substance. From now on, when we say the chemical formula of a substance made up of molecules, we mean its molecular formula.

The chlorine molecule

N2 Examination questions often ask about halogens.
- Halogens are non-metals.
- Halogens exist as diatomic molecules.
- Halogens form covalent compounds with non-metals.
- Halogens form ionic compounds with metals.

A chlorine atom has an electronic arrangement of 2,8,7. Each chlorine atom can obtain the electronic arrangement of an argon atom (2,8,8) by sharing one of its outermost shell electrons with another chlorine atom (Fig. 8.3). N2

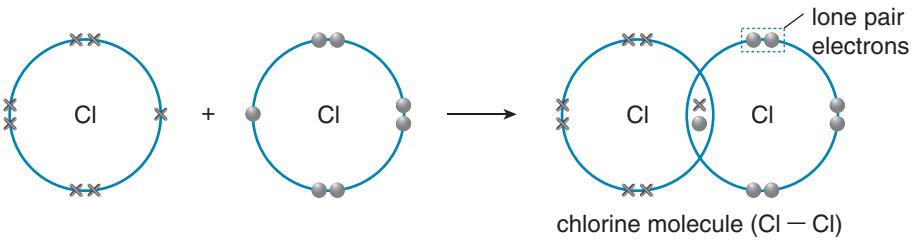


Fig. 8.3 Formation of covalent bond in a chlorine molecule (only electrons in the outermost shell are shown)

bond pair electrons 鍵合電子對　diatomic molecule 雙原子分子　monoatomic molecule 單原子分子
molecular formula 分子式

In a chlorine molecule (Fig. 8.4), each chlorine atom has three electron pairs in the outermost shell that are not involved in the bonding.

✓ The pair of electrons in the outermost shell that is not involved in bonding is called **lone pair electrons**.

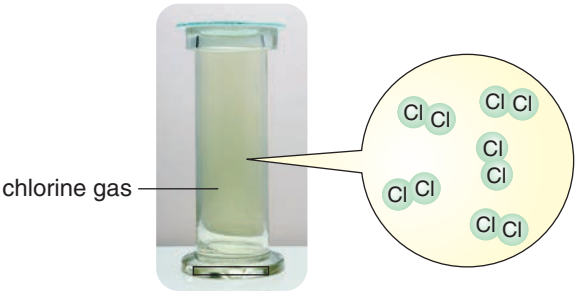
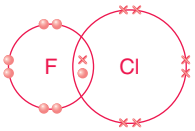


Fig. 8.4 Chlorine exists as diatomic molecules

The hydrogen chloride molecule

N3 A hydrogen atom shares electrons with one fluorine atom in a similar way.

N4 Examination questions may ask students to draw electron diagrams of compounds formed from two halogens. e.g.



(Only electrons in the outermost shells are shown.)

A hydrogen atom has an electronic arrangement of 1 while a chlorine atom has an electronic arrangement of 2,8,7. In order to obtain the electronic arrangements of atoms of noble gases, one hydrogen N3 atom forms a single covalent bond with one chlorine atom (Fig. 8.5). A hydrogen chloride molecule results. The chemical formula of N4 hydrogen chloride is HCl.

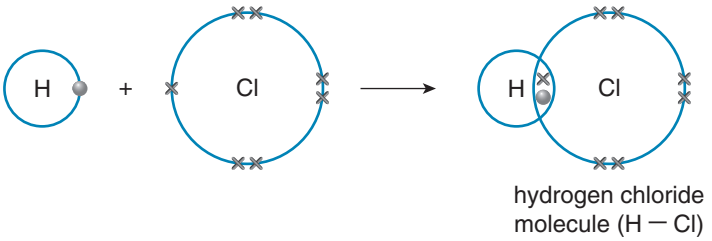


Fig. 8.5 Formation of covalent bond in a hydrogen chloride molecule (only electrons in the outermost shells are shown)

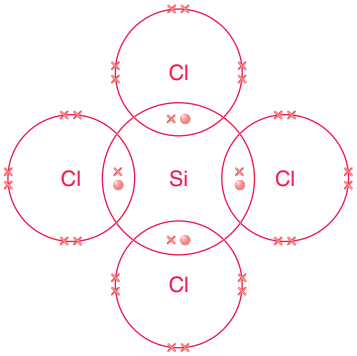
Hydrogen chloride is a **covalent compound**, as a covalent bond holds the two atoms together.

lone pair electrons 孤電子對　　covalent compound 共價化合物

The tetrachloromethane molecule

A carbon atom has an electronic arrangement of 2,4 while a chlorine atom has an electronic arrangement of 2,8,7. In order to obtain the electronic arrangements of atoms of noble gases, one carbon atom forms N5 a single covalent bond with each of four chlorine atoms (Fig. 8.6). A **tetrachloromethane molecule** results. The chemical formula of tetrachloromethane is CCl_4.

N5 A silicon atom shares electrons with four chlorine atoms in a similar way.



(Only electrons in the outermost shells are shown.)

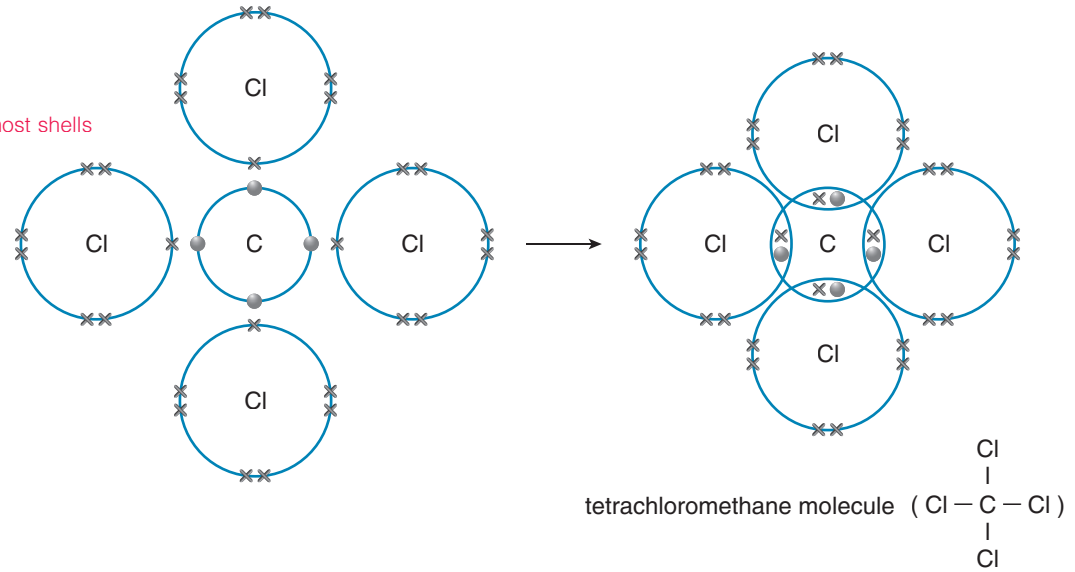
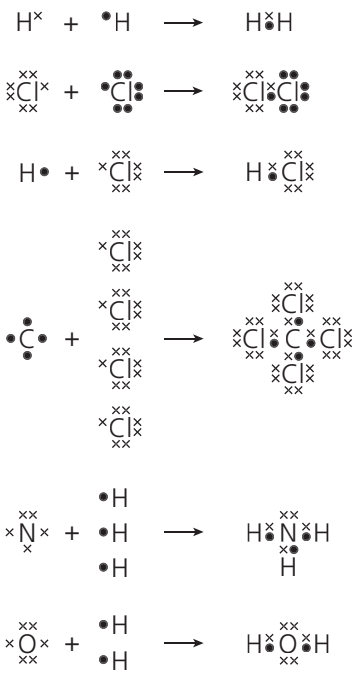


Fig. 8.6 Formation of covalent bonds in a tetrachloromethane molecule (only electrons in the outermost shells are shown)

N6 Refer to page T36.

N7 Refer to page T36.

We can simplify Figs. 8.1, 8.3, 8.5–8.8 as below:



The ammonia molecule

A nitrogen atom has an electronic arrangement of 2,5 while a hydrogen atom has an electronic arrangement of 1. In order to obtain the electronic arrangements of atoms of noble gases, one nitrogen atom N6 forms a single covalent bond with each of three hydrogen atoms N7 (Fig. 8.7). An **ammonia molecule** results. The chemical formula of ammonia is NH_3.

The nitrogen atom in an ammonia molecule has three bond pairs and one lone pair.

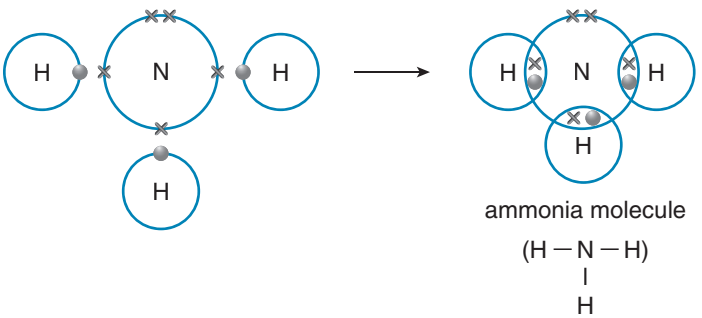


Fig. 8.7 Formation of covalent bonds in an ammonia molecule (only electrons in the outermost shells are shown)

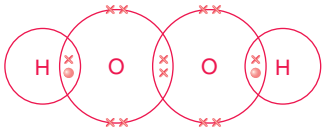
tetrachloromethane molecule 四氯甲烷分子　　ammonia molecule 氨分子

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The water molecule

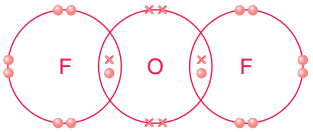
An oxygen atom has an electronic arrangement of 2,6 while a hydrogen atom has an electronic arrangement of 1. In order to obtain the electronic arrangements of atoms of noble gases, one oxygen atom forms a single covalent bond with each of two hydrogen atoms (Fig. 8.8). A water molecule results. The chemical formula of water is H_2O.

N8 Hydrogen peroxide (H_2O_2) is an oxide of hydrogen.



(Only electrons in the outermost shells are shown.)

N9 An oxygen atom shares electrons with two fluorine atoms in a similar way.



(Only electrons in the outermost shells are shown.)

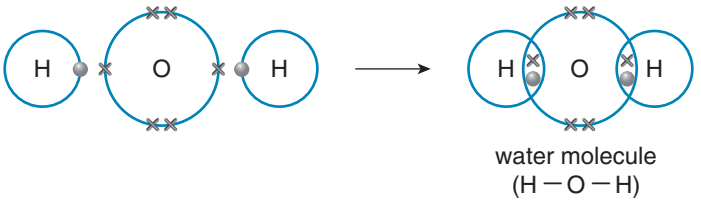


Fig. 8.8 Formation of covalent bonds in a water molecule (only electrons in the outermost shells are shown)

How many single covalent bonds can an atom form?

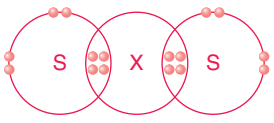
The position of an element in the periodic table tells us how many electrons its atom needs in order to obtain the electronic arrangement of the atom of a noble gas. Thus, it tells us how many single covalent bonds the atom can form (Fig. 8.9). For example, a nitrogen atom (Group V) needs another three electrons, so it can form three single covalent bonds (e.g. in ammonia, NH_3). A carbon atom (Group IV) needs another four electrons, so it can form four single covalent bonds (e.g. in tetrachloromethane, CCl_4).

Group in the periodic table to which the element belongs	I	II	III	IV	V	VI	VII	0
Number of single covalent bond(s) its atom can form	—	—	—	4	3	2	1	— N10

Fig. 8.9 The position of an element in the periodic table and the number of single covalent bond(s) its atom can form N11

N10 Refer to page T37.

N11 Examination questions may ask students to deduce the number of electrons in the outermost shell of an atom from the electron diagram of a compound.
e.g.



There are 4 electrons in the outermost shell of an atom of X.

Practice 8.1

1 The following table lists the atomic numbers of four elements.

Element	Atomic number
W	6
X	9
Y	15
Z	17

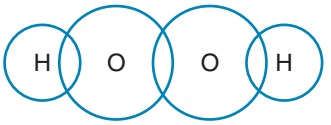
Use an electron diagram to show the sharing of electrons in the compound formed from each pair of elements. Show electrons in the *outermost shells* only.

a) W and X

b) X and Z

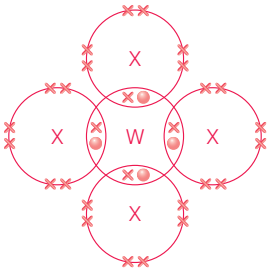
c) Y and Z

2 Hydrogen peroxide (H_2O_2) is an oxide of hydrogen. Complete the diagram below for the hydrogen peroxide molecule using dots or crosses to represent electrons. Show electrons in the *outermost shells* only.

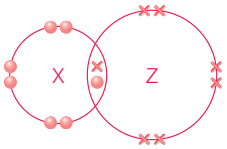


Practice 8.1

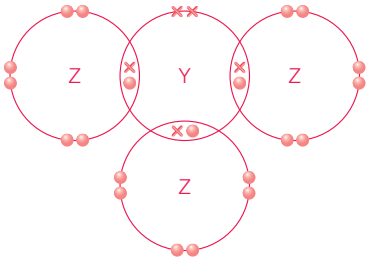
1 a)



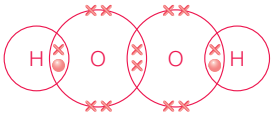
b)



c)



2



8.2 Covalent molecules with multiple covalent bonds

There are many examples where non-metallic atoms must form multiple covalent bonds to obtain stable electronic arrangements.

The oxygen molecule

An oxygen atom has an electronic arrangement of 2,6. Each oxygen atom can obtain the electronic arrangement of a neon atom (2,8) by sharing two of its electrons with another oxygen atom (Fig. 8.10).

Since the oxygen atoms share two pairs of electrons, the bond between them is called a **double bond**. An oxygen molecule can be shown as O=O.

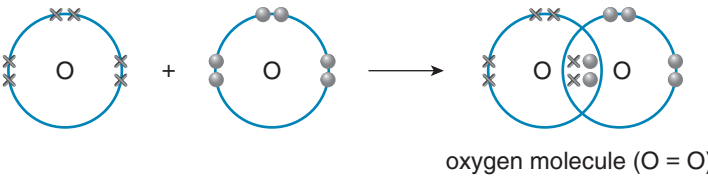


Fig. 8.10 Formation of covalent bond in an oxygen molecule (only electrons in the outermost shell are shown)

double bond 雙鍵

The nitrogen molecule

A nitrogen atom has an electronic arrangement of 2,5. Each nitrogen atom can obtain the electronic arrangement of a neon atom (2,8) by sharing three of its electrons with another nitrogen atom (Fig. 8.11).

Since the nitrogen atoms share three pairs of electrons, the bond between them is called a **triple bond**. A nitrogen molecule can be shown as N≡N.

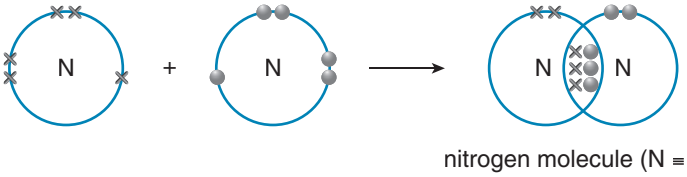
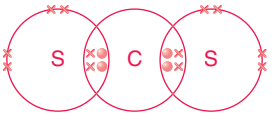


Fig. 8.11 Formation of covalent bond in a nitrogen molecule (only electrons in the outermost shell are shown)

N12 A carbon atom shares electrons with two sulphur atoms in a similar way.



(Only electrons in the outermost shells are shown.)

The carbon dioxide molecule

A carbon atom has an electronic arrangement of 2,4 while an oxygen atom has an electronic arrangement of 2,6. In order to obtain the electronic arrangements of atoms of noble gases, one carbon atom N12 forms a double bond with each of two oxygen atoms (Fig. 8.12). A carbon dioxide molecule results. The chemical formula of carbon dioxide is CO_2.

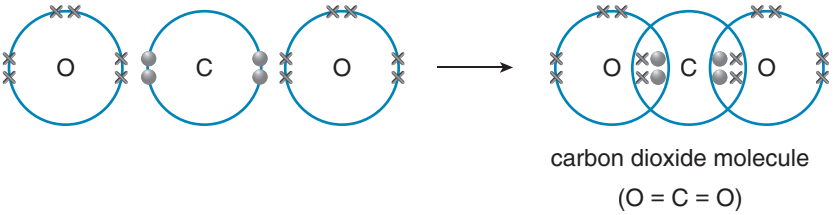
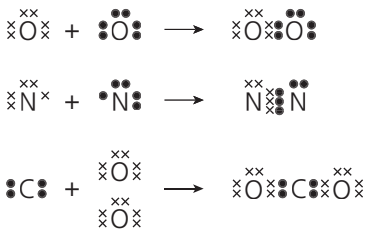


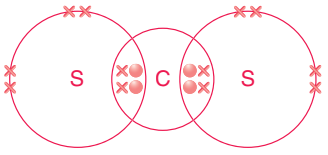
Fig. 8.12 Formation of covalent bonds in a carbon dioxide molecule (only electrons in the outermost shells are shown)

We can simplify Figs. 8.10–8.12 as below:

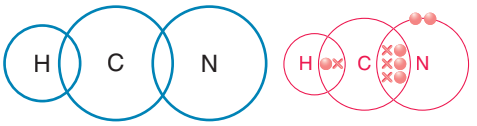


Practice 8.2

1 Use an electron diagram to show the sharing of electrons in the compound formed from carbon and sulphur. Show electrons in the *outermost shells* only.



2 Complete the diagram below for a HCN molecule using dots or crosses to represent electrons. Show electrons in the *outermost shells* only.



triple bond 三鍵

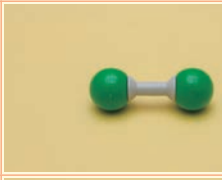
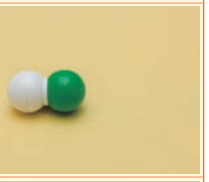
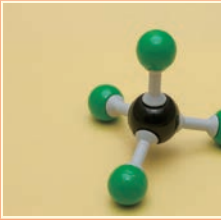
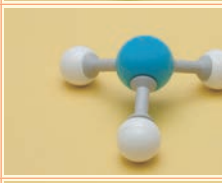
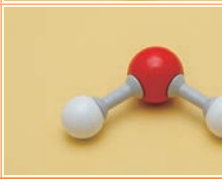
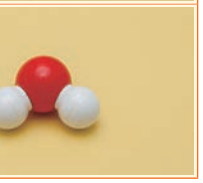
8.3 Molecular models

8.1
Building models of some covalent molecules.

We can build models of the molecules. The simplest models are **ball-and-stick models**. To get a more accurate idea of how close together the atoms in a molecule are, we use **space-filling models**. Table 8.1a and Table 8.1b show the molecular models of molecules with single covalent bonds and multiple covalent bonds respectively.

Table 8.1a

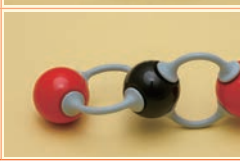
Molecular models of molecules with single covalent bonds

Molecule	Ball-and-stick model	Space-filling model
Hydrogen (H — H)		
Chlorine (Cl — Cl)		
Hydrogen chloride (H — Cl)		
Tetrachloromethane Cl (Cl — C — Cl) Cl		
Ammonia (H — N — H) H		
Water (H — O — H)		

molecular model 分子模型　　ball-and-stick model 球棒模型　　space-filling model 填空模型

Table 8.1b

Molecular models of molecules with multiple covalent bonds

Molecule	Ball-and-stick model	Space-filling model
Oxygen (O = O)		
Nitrogen (N ≡ N)		
Carbon dioxide (O = C = O)		

8.4 Writing chemical formulae of covalent compounds

Suppose we need to work out the chemical formula of the covalent compound formed from each of the following pairs of elements:

- nitrogen and fluorine;
- silicon and oxygen.

Refer to Table 8.2 for the steps involved.

The compound NF_3 formed from nitrogen and fluorine is called nitrogen trifluoride.

Table 8.2

Steps for working out the chemical formulae of covalent compounds

Step	Compound formed from nitrogen and fluorine	Compound formed from silicon and oxygen
1 Write down the electronic arrangements of the atoms involved.	N 2,5 F 2,7	Si 2,8,4 O 2,6
2 Decide the number of electrons each atom needs to obtain a stable electronic arrangement. Write down the number on the top of each atom.	A nitrogen atom needs 3 electrons, while a fluorine atom needs 1 electron. 3 N 1 F	A silicon atom needs 4 electrons, while an oxygen atom needs 2 electrons. 4 Si 2 O
3 Decide the number of each type of atom in one molecule (cross multiply the numbers and the symbols).	3 N, 1 F → = N, = F_3	4 Si, 2 O → = Si_2, = O_4
4 Combine the symbols and simplify the ratio if necessary.	NF_3	SiO_2 (Simplify the ratio of 2 : 4 to 1 : 2.)

Reference Website
Refer to animations of chemical bonds in compounds formed between different elements:
http://group.chem.iastate.edu/Greenbowe/sections/projectfolder/flashfiles/reaction/bonding1.html
(accessed July 2014)

8.5 Predicting the formation of ionic and covalent compounds

We can predict whether the compound formed is ionic or covalent from the information about the elements that combine.

- ✓ When a metal combines with a non-metal, an ionic compound forms.
- ✓ When non-metals combine, a covalent compound forms.

We will discuss these structures in Unit 9.

Fig. 8.13 summarizes the constituent particles of metallic elements, ionic compounds, covalent compounds and non-metallic elements.

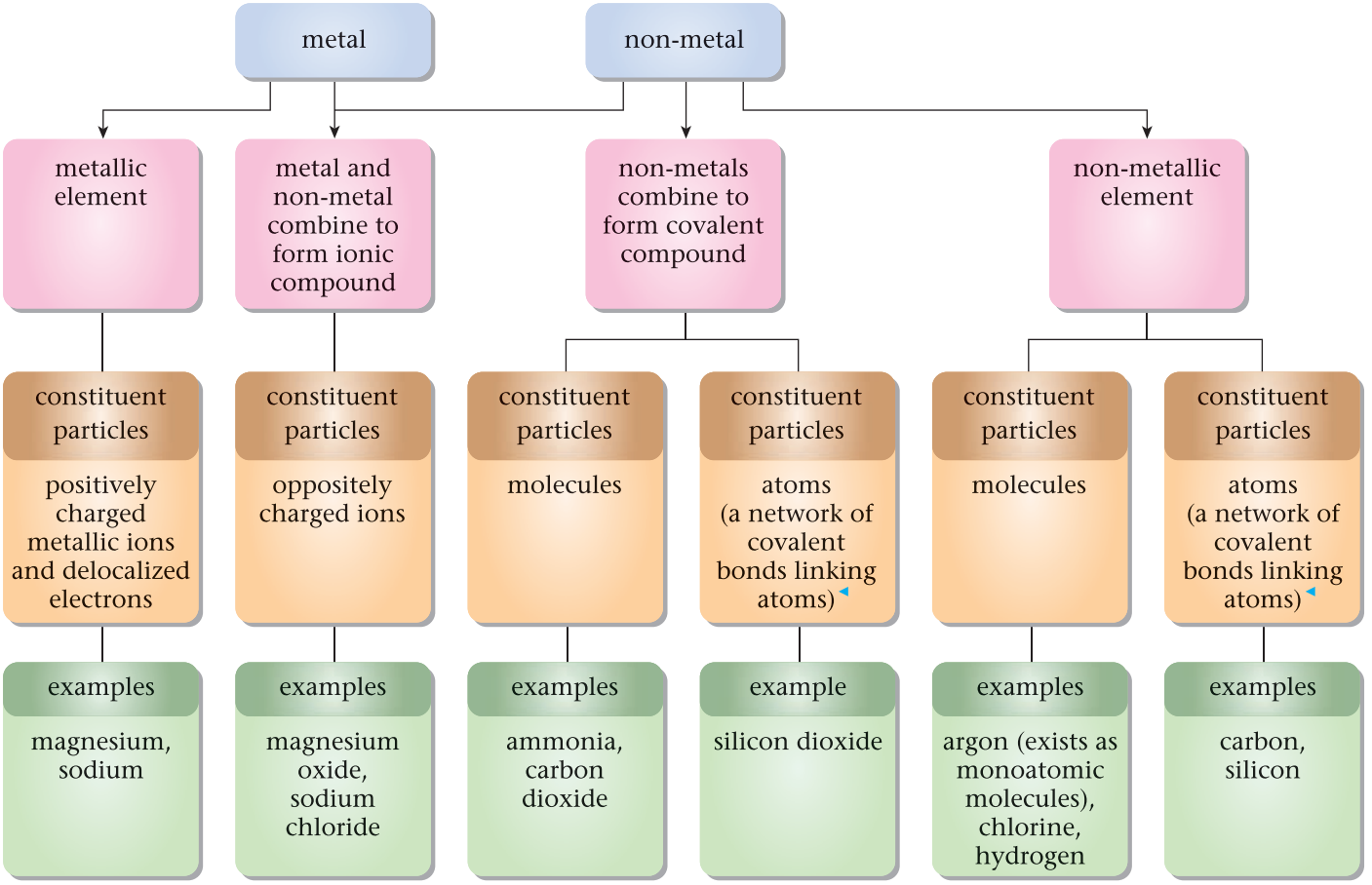


Fig. 8.13 Constituent particles in metallic elements, ionic compounds, covalent compounds and non-metallic elements

Example 8.1

Q X, Y and Z are three consecutive elements in the periodic table, with the sum of their atomic numbers equal to 30. X forms a stable anion X^- while Z forms a stable cation Z^+.

a) What are elements X, Y and Z?

b) P is a compound formed from element X and oxygen.

i) Predict the type of compound (ionic or covalent) formed. Explain your answer.

ii) Draw an electron diagram of P, showing electrons in the *outermost shells* only.

iii) Give the name of P.

c) Q is a compound formed from element Z and oxygen.

i) Predict the type of compound (ionic or covalent) formed. Explain your answer.

ii) Draw an electron diagram of Q, showing electrons in the *outermost shells* only.

iii) Give the name of Q.

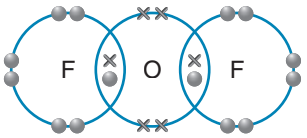
A a) X: fluorine

Y: neon

Z: sodium

b) i) When fluorine and oxygen (non-metals) combine, a covalent compound is formed.

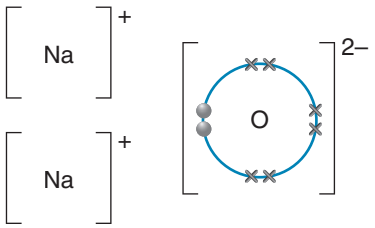
ii)



iii) Oxygen difluoride

c) i) When sodium (a metal) combines with oxygen (a non-metal), an ionic compound is formed.

ii)



iii) Sodium oxide

Practice 8.3

W, X, Y and Z are four different elements. The electronic arrangements of their atoms are listed below.

Element	Electronic arrangement of atom
W	2,5
X	2,8,7
Y	2,8,8,2
Z	*p*,*q*,*r*,8,2

a) Elements Y and Z belong to the same group.

i) What are the numerical values for *p*, *q* and *r* in the electronic arrangement of an atom of Z?

ii) Predict which of the elements, Y or Z, would react more vigorously with water. Explain your answer.

b) P is a compound formed from elements W and X.

i) Predict the type of compound (ionic or covalent) formed.

ii) Draw an electron diagram of P, showing electrons in the *outermost shells* only.

iii) Give the chemical formula and name of P.

c) Q is a compound formed from elements W and Y.

i) Predict the type of compound (ionic or covalent) formed.

ii) Draw an electron diagram of Q, showing electrons in the *outermost shells* only.

iii) Give the chemical formula and name of Q.

Practice 8.3

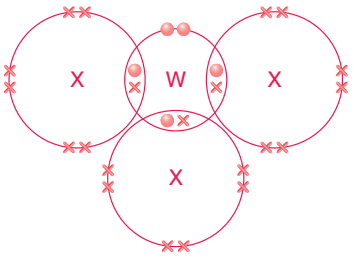
a) i) $p = 2$, $q = 8$, $r = 18$

ii) Z

The reactivity of Group II metals increases down the group.

b) i) Covalent compound

ii)

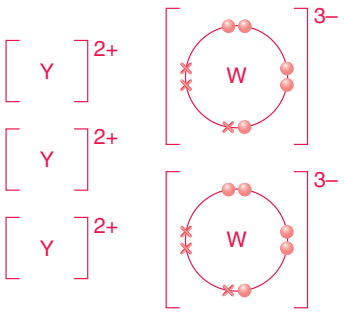


iii) NCl_3

Nitrogen trichloride

c) i) Ionic compound

ii)



iii) Ca_3N_2

Calcium nitride

8.6 Dative covalent bonding

In a covalent bond, two atoms can share one, two or three pairs of electrons to form different covalent bonds. Usually each atom supplies one electron to make up the pair. Sometimes, however, both of the electrons come from the same atom. This atom is called the **donor atom**, and the two electrons it provides come from a lone pair on that atom. The other atom in the bond is called the **acceptor atom**. This kind of bonding is called **dative covalent bonding**. The word 'dative' means 'giving'.

✓ A dative covalent bond is a covalent bond in which the bond pair electrons are provided by the same atom.

donor atom 給體原子　　acceptor atom 受體原子　　dative covalent bonding 配位共價鍵

Ammonium ion (NH_4^+)

The hydrogen ion is a 'bare' proton with no electrons at all and thus electron-deficient.

Ammonia forms a dative covalent bond when it combines with a hydrogen ion to make an ammonium ion. The nitrogen atom in the ammonia molecule has a lone pair of electrons. This lone pair provides both the electrons for bonding with the hydrogen ion. The resulting ammonium ion carries one positive charge (Fig. 8.14a).

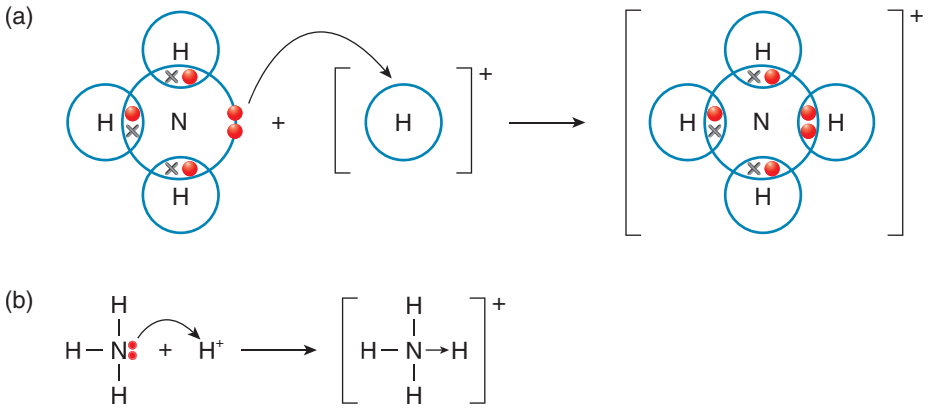


Fig. 8.14 Formation of a dative covalent bond in an ammonium ion (a) electron diagram (b) single line representation

An arrow is used to represent the dative covalent bond in the NH_4^+ ion (Fig. 8.14b). The arrow points from the atom donating the electron pair to the atom accepting them.

For dative covalent bonding to occur, we need

- one atom having a lone pair of electrons;
- a second particle having an unfilled electron shell to accept the lone pair, in other words an electron-deficient species.

Hydroxonium ion or hydronium ion (H_3O^+)

We will further discuss the H_3O^+ ion in Topic 4 Acids and Bases.

A dative covalent bond is also formed when a water molecule combines with a hydrogen ion to form a *hydroxonium ion* or *hydronium ion* (H_3O^+). The oxygen atom in the water molecule has two lone pairs of electrons. One lone pair provides both the electrons for bonding with the hydrogen ion. The resulting ion carries one positive charge (Fig. 8.15a–b).

hydroxonium ion or hydronium ion 水合氫離子

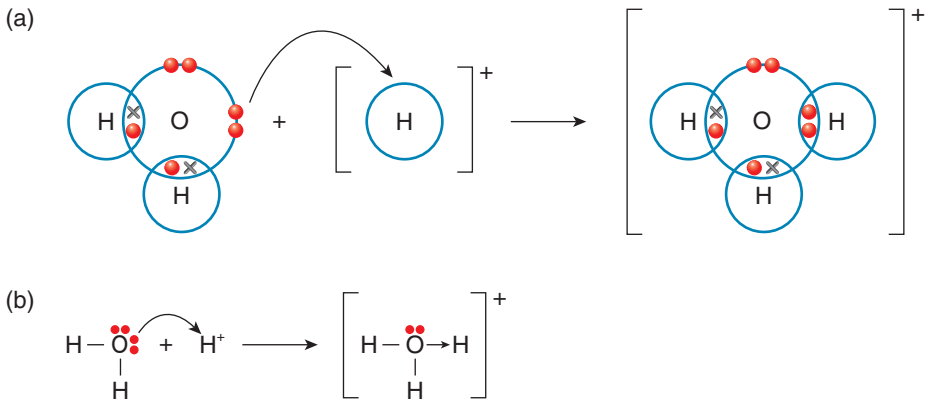


Fig. 8.15 Formation of a dative covalent bond in a hydroxonium ion or hydronium ion (a) electron diagram (b) single line representation

NH_3BF_3 molecule

N13 BF_3 molecule and F^- ion combine to form BF_4^- ion.
$BF_3 + F^- \longrightarrow BF_4^-$
The F^- ion provides both the bonding electrons to the boron atom.

In a **boron trifluoride molecule**, the boron atom has only 6 outermost shell electrons and it needs two more electrons to attain the octet structure. N13

When gaseous ammonia (NH_3) and gaseous boron trifluoride (BF_3) react together, a solid NH_3BF_3 is formed. The lone pair of the nitrogen atom provides both the bonding electrons to the boron atom when ammonia bonds with boron trifluoride. As a result, the boron atom attains an octet structure (Fig. 8.16a–b).

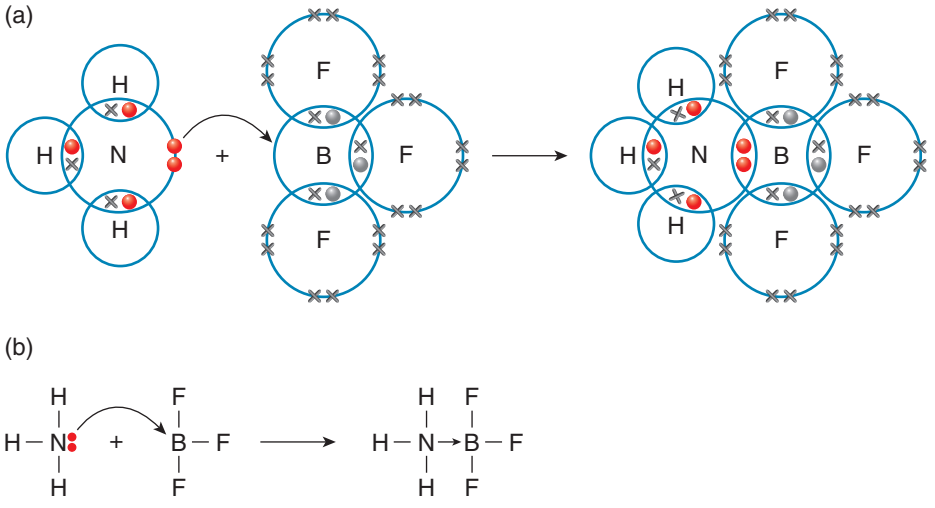


Fig. 8.16 Formation of a dative covalent bond in a NH_3BF_3 molecule (a) electron diagram (b) single line representation

boron trifluoride molecule 三氟化硼分子

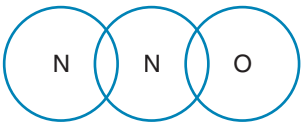
Practice 8.4

1 Nitrogen forms an oxide N_2O. The bonding in N_2O can be represented as below:

N≡N→O

a) What does the arrow (→) represent?

b) Complete the diagram below for a N_2O molecule using dots or crosses to represent electrons. Show electrons in the *outermost shells* only.

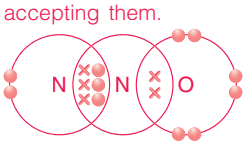


2 Put tick(s) '✔' to show the bond(s) existing in each compound.

Compound	Ionic bond	Covalent bond
Ammonium chloride	✔	✔
Magnesium oxide	✔	
Hydrogen sulphide		✔

1 a) The arrow is used to represent the dative covalent bond in N_2O. The arrow points from the atom donating the electron pair to the atom accepting them.

b)



Relative molecular mass has no unit.

8.7 Relative molecular mass and formula mass

Relative molecular mass

Just as the relative atomic mass is used to describe the relative masses of atoms, the **relative molecular mass** is used to describe the relative masses of molecules.

✓ Relative molecular mass of an element or compound = sum of relative atomic masses of all atoms present in one molecule of the element or compound

Chlorine exists as diatomic molecules. Each chlorine molecule contains two chlorine atoms.

Relative molecular mass of chlorine
= 2 x relative atomic mass of Cl
= 2 x 35.5
= 71.0

Ammonia is a covalent compound. Each ammonia molecule (NH_3) contains one nitrogen atom and three hydrogen atoms.

Relative molecular mass of ammonia
= relative atomic mass of N + 3 x relative atomic mass of H
= 14.0 + 3 x 1.0
= 17.0

relative molecular mass 相對分子質量

Example 8.2

Q Calculate the relative molecular mass of table sugar, sucrose, which has the molecular formula $C_{12}H_{22}O_{11}$.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

A Relative molecular mass of sucrose
= 12 x relative atomic mass of C + 22 x relative atomic mass of H + 11 x relative atomic mass of O
= 12 x 12.0 + 22 x 1.0 + 11 x 16.0
= 342.0

∴ the relative molecular mass of sucrose is 342.0.

Formula mass

Formula mass has no unit.

Ionic compounds consist of cations and anions. Since ionic compounds do not contain molecules, we use **formula mass** to describe the relative masses of ionic compounds.

The formula unit is the simplest unit of a pure substance.

✓ Formula mass of an ionic compound
= sum of the relative atomic masses of all atoms making up one formula unit of the compound

Potassium chloride (KCl) is an ionic compound. Every formula unit of potassium chloride contains one potassium ion and one chloride ion.

Formula mass of potassium chloride (KCl)
= relative atomic mass of K + relative atomic mass of Cl
= 39.1 + 35.5
= 74.6

Ammonium sulphate ($(NH_4)_2SO_4$) is an ionic compound. Every formula unit of ammonium sulphate contains two ammonium ions and one sulphate ion. An ammonium ion consists of one nitrogen atom and four hydrogen atoms. A sulphate ion consists of one sulphur atom and four oxygen atoms.

Formula mass of ammonium sulphate ($(NH_4)_2SO_4$)
= 2 x (relative atomic mass of N + 4 x relative atomic mass of H)
+ relative atomic mass of S + 4 x relative atomic mass of O
= 2 x (14.0 + 4 x 1.0) + 32.1 + 4 x 16.0
= 132.1

formula mass 式量

Practice 8.5

1 Calculate the formula mass of calcium phosphate, $Ca_3(PO_4)_2$, commonly called rock phosphate.

(Relative atomic masses: O = 16.0, P = 31.0, Ca = 40.1)

2 Calculate the formula mass or relative molecular mass of each of the following substances.

Substance	Chemical formula	Relative atomic mass	Formula mass / relative molecular mass
Water	H_2O	H = 1.0 O = 16.0	18.0
Sodium carbonate	Na_2CO_3	C = 12.0 O = 16.0 Na = 23.0	106.0
Magnesium hydroxide	$Mg(OH)_2$	H = 1.0 O = 16.0 Mg = 24.3	58.3
Iron(III) sulphate	$Fe_2(SO_4)_3$	O = 16.0 S = 32.1 Fe = 55.8	399.9

3 The table below lists some information of two elements, X and Y:

Element	Atomic number	Relative atomic mass
X	8	16.0
Y	13	27.0

a) Assume elements X and Y can form an ionic compound. Deduce the chemical formula of this compound.

b) Calculate the formula mass of the compound formed between elements X and Y.

1 Formula mass of calcium phosphate
= 3 x relative atomic mass of Ca + 2 x (relative atomic mass of P + 4 x relative atomic mass of O)
= 3 x 40.1 + 2 x (31.0 + 4 x 16.0)
= 310.3
∴ the formula mass of calcium phosphate is 310.3.

3 a) Element X is oxygen and Y is aluminium.
The aluminium ion (Al^{3+}) carries 3 positive charges while the oxide ion (O^{2-}) carries 2 negative charges. For electrical neutrality, we need three oxide ions for every two aluminium ions. So, the chemical formula of the compound is Al_2O_3 (or Y_2X_3).
b) Formula mass of the compound
= 2 x relative atomic mass of Y + 3 x relative atomic mass of X
= 2 x 27.0 + 3 x 16.0
= 102.0

Have you mastered?

Key terms

covalent bond 共價鍵 104 molecule 分子 104 single bond 單鍵 104

bond pair electrons 鍵合電子對 105 diatomic molecule 雙原子分子 105

monoatomic molecule 單原子分子 105 molecular formula 分子式 105

lone pair electrons 孤電子對 106 covalent compound 共價化合物 106

tetrachloromethane molecule 四氯甲烷分子 107 ammonia molecule 氨分子 107

double bond 雙鍵 109 triple bond 三鍵 110 molecular model 分子模型 111

ball-and-stick model 球棒模型 111 space-filling model 填空模型 111 donor atom 給體原子 115

acceptor atom 受體原子 115 dative covalent bond 配位共價鍵 115 ammonium ion 銨離子 116

hydroxonium ion or hydronium ion 水合氫離子 116 boron trifluoride molecule 三氟化硼分子 117

relative molecular mass 相對分子質量 118 formula mass 式量 119

Checklist

After studying this unit, you should be able to

- ☐ describe the formation of a covalent bond;
- ☐ describe the formation of single, double and triple bonds using electron diagrams;
- ☐ interpret chemical formulae of covalent compounds in terms of the elements present and the ratio of their atoms;
- ☐ write the names and formulae of covalent compounds based on their component atoms;
- ☐ predict the type of compound (ionic or covalent) formed from the information about the elements that combine;
- ☐ describe the formation of a dative covalent bond;

☐ define and distinguish between relative molecular mass and formula mass;

☐ perform calculations related to relative molecular masses and formula masses.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

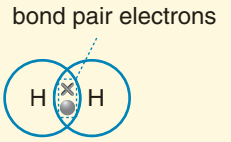
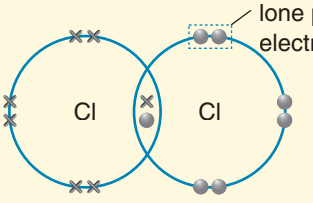
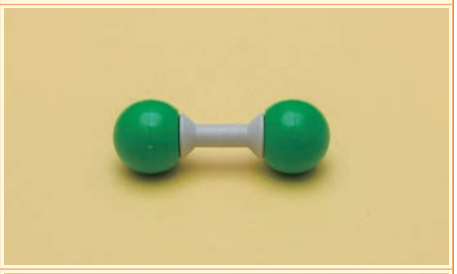
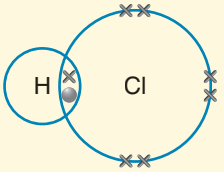
1 A covalent bond is formed when one or more pairs of outermost shell electrons are shared between two atoms.

2 A covalent bond is the strong electrostatic forces of attraction between the shared electrons and the two positively charged nuclei of the bonded atoms.

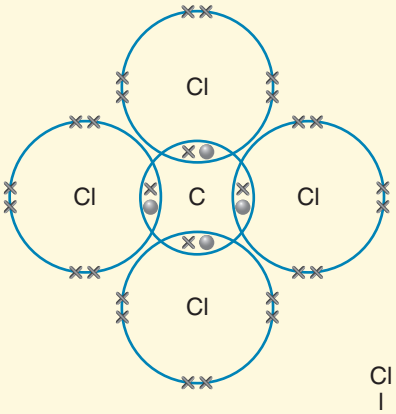
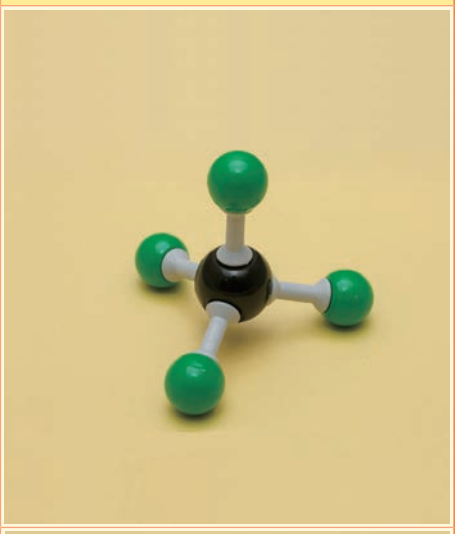
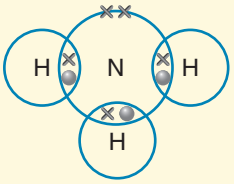
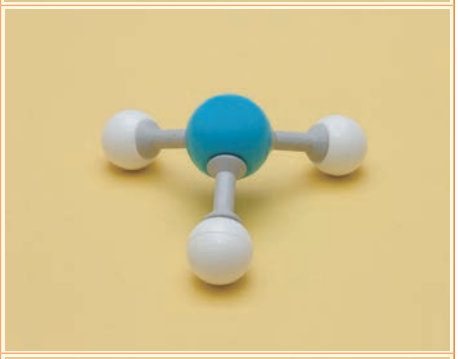
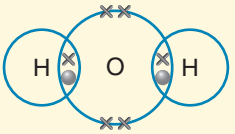
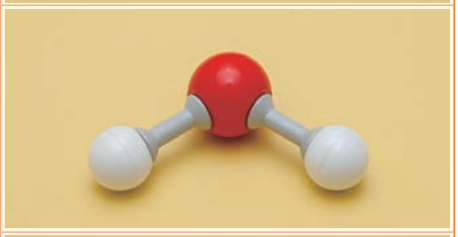
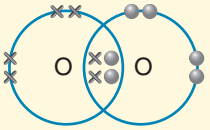
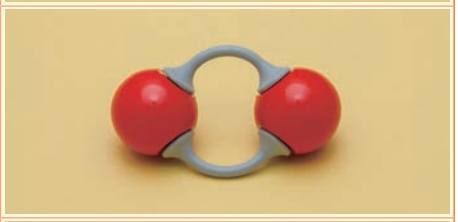
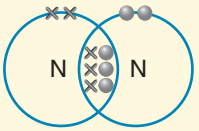
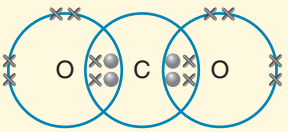
3 Non-metallic atoms can join together to form groups called molecules.

4 Atoms of different non-metals share electrons with each other to form covalent compounds.

5 The table below shows the electron diagrams and models of molecules of some common non-metals and covalent compounds.

Molecule	Electron diagram	Ball-and-stick model
Hydrogen	 hydrogen molecule (H — H)	
Chlorine	 chlorine molecule (Cl — Cl)	
Hydrogen chloride	 hydrogen chloride molecule (H — Cl)	

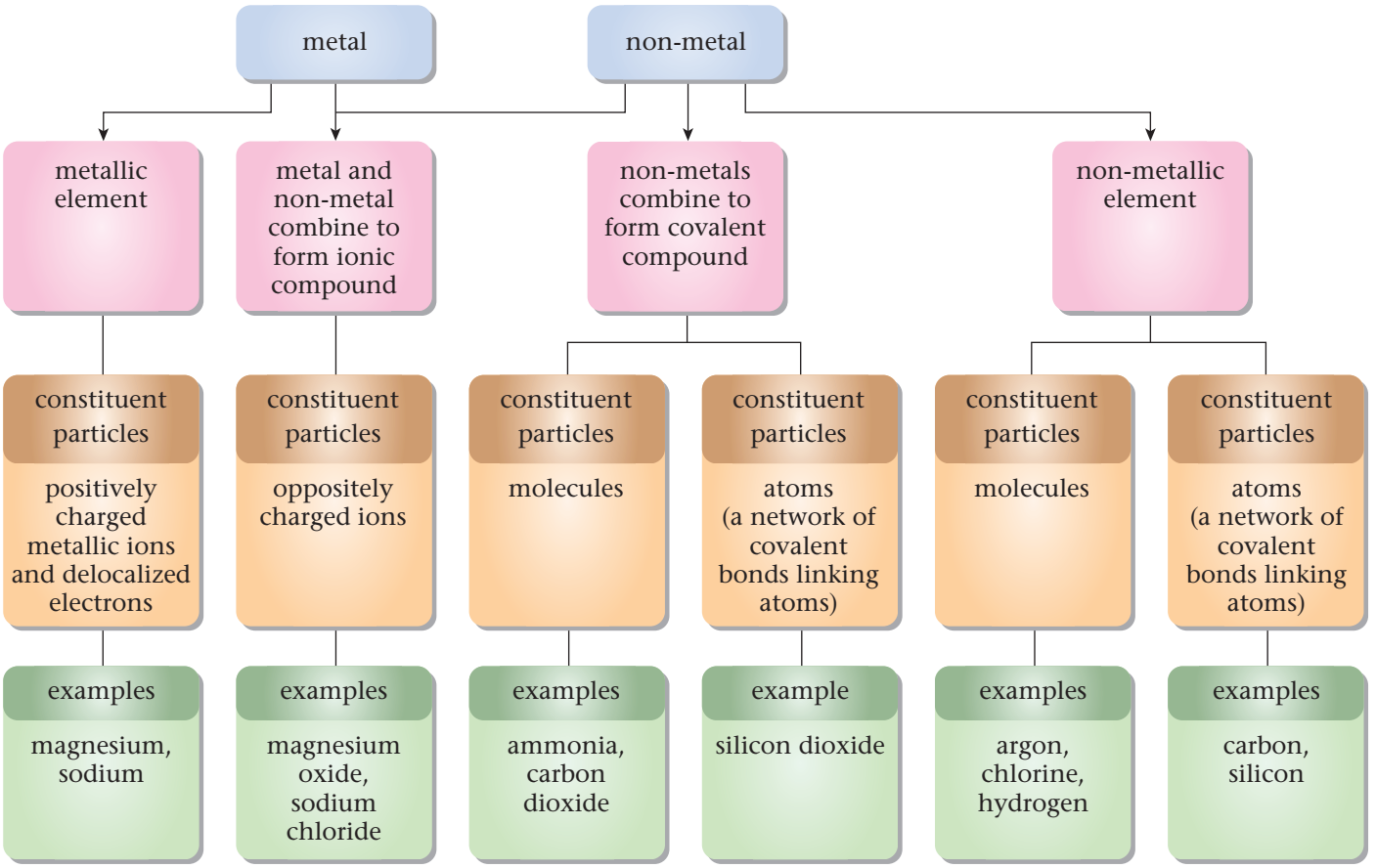
Continued on next page

Molecule	Electron diagram	Ball-and-stick model
Tetrachloromethane	 tetrachloromethane molecule (Cl — C — Cl)	
Ammonia	 ammonia molecule (H — N — H)	
Water	 water molecule (H — O — H)	
Oxygen	 oxygen molecule (O = O)	
Nitrogen	 nitrogen molecule (N ≡ N)	
Carbon dioxide	 carbon dioxide molecule (O = C = O)	

Have you mastered?

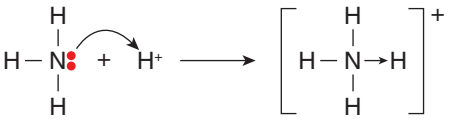
6 a) When a metal combines with a non-metal, an ionic compound forms.

b) When non-metals combine, a covalent compound forms.



7 A dative covalent bond is a covalent bond in which the bond pair electrons are provided by the same atom.

The following diagram shows the formation of a dative covalent bond in an ammonium ion.



8 Relative molecular mass of an element or compound = sum of relative atomic masses of all atoms present in one molecule of the element or compound

9 Formula mass of an ionic compound = sum of the relative atomic masses of all atoms making up one formula unit of the compound

Have you mastered?

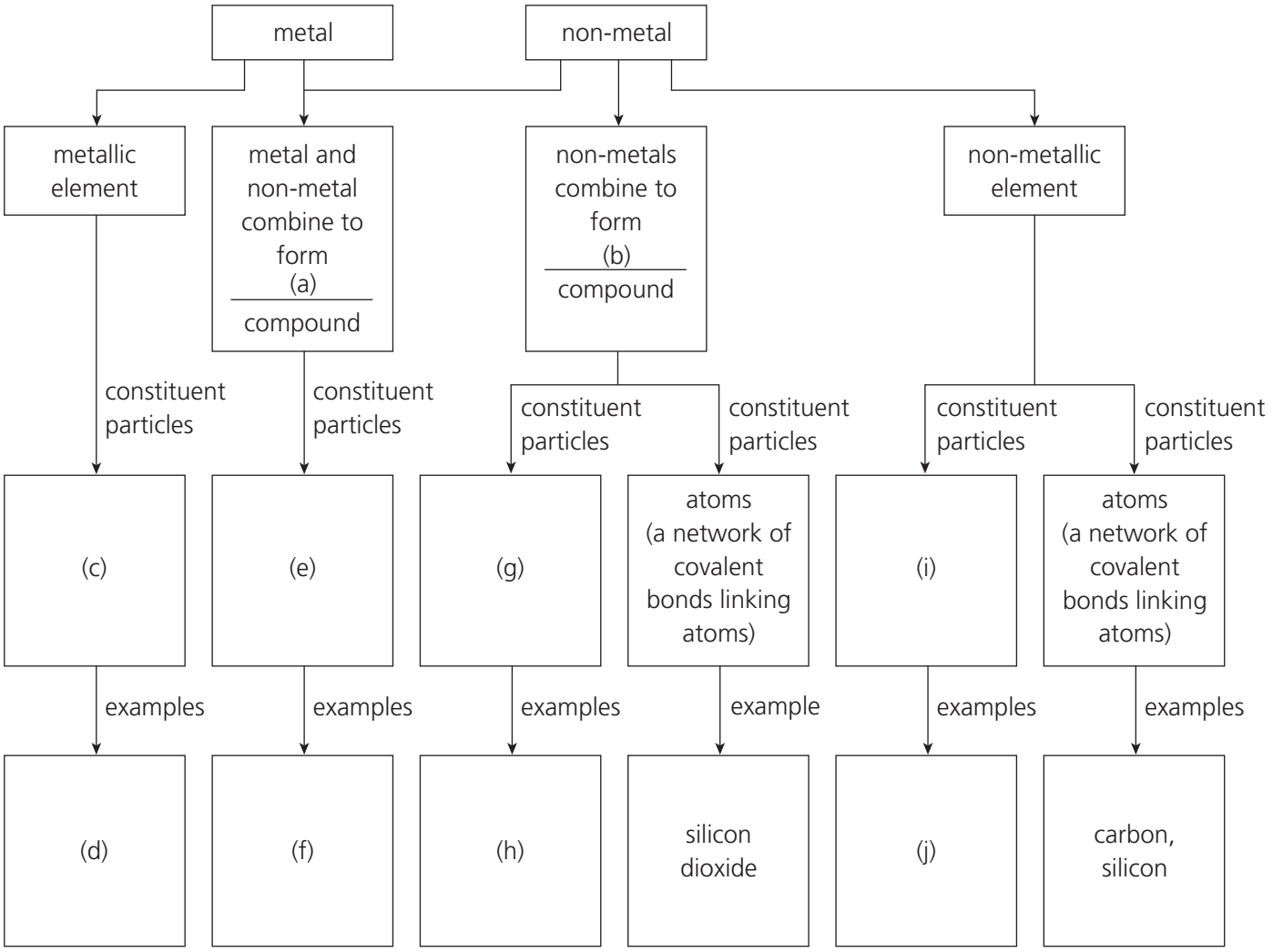
Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the concept map using the following words:

ammonia, carbon dioxide; argon, chlorine, hydrogen; covalent; ionic; magnesium, sodium; magnesium oxide, sodium chloride; molecules (2 times); oppositely charged ions; positively charged metallic ions and delocalized electrons



a) ionic
b) covalent
c) positively charged metallic ions and delocalized electrons
d) magnesium, sodium
e) oppositely charged ions
f) magnesium oxide, sodium chloride
g) molecules
h) ammonia, carbon dioxide
i) molecules
j) argon, chlorine, hydrogen

2 Classify each of the following compounds as ionic or covalent.

Compound	Ionic / Covalent
Copper(II) chloride	Ionic
Tetrachloromethane	Covalent
Water	Covalent
Sodium oxide	Ionic

3 Here are the names of four elements in the periodic table.

bromine, iodine, potassium, lithium

Choose from these names to answer the following questions.

a) Which of these elements

i) exist as diatomic molecules?
Bromine and iodine

ii) react with water to make hydrogen gas?
Potassium and lithium

iii) has a melting point below room temperature?
Bromine

b) Which two of these elements will react together most violently? Potassium and bromine

(OCR GCSE 21st Century Science (Higher Tier), Chem. A, Unit 2, Jun. 2008, 3(c))

Unit Exercise

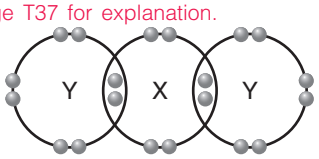
4 Complete the following table.

Substance	Chemical formula	Relative atomic mass	Formula mass / relative molecular mass
Nitrogen dioxide	NO_2	N = 14.0 O = 16.0	46.0
Sodium sulphate	Na_2SO_4	O = 16.0 Na = 23.0 S = 32.1	142.1
Iron(III) nitrate	$Fe(NO_3)_3$	N = 14.0 O = 16.0 Fe = 55.8	241.8
Ammonium phosphate	$(NH_4)_3PO_4$	H = 1.0 N = 14.0 O = 16.0 P = 31.0	149.0

Part II Multiple choice questions

5 The electron diagram of a compound formed between element X and element Y is shown below:

Refer to page T37 for explanation.



(Only electrons in the *outermost shells* are shown.)

Which of the following combinations concerning X and Y is correct?

	X	Y
A	carbon	sulphur
B	silicon	oxygen
C	oxygen	potassium
(D)	sulphur	fluorine

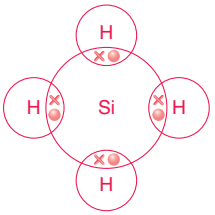
6 Which of the following pairs of elements would form a covalent compound?

A Helium and hydrogen
B Iron and fluorine
C Neon and nitrogen
(D) Phosphorus and chlorine

Phosphorus and chlorine are non-metals. They combine to form a covalent compound.

7 Which of the following molecules does NOT contain any lone pair of electrons?

A F_2
B NH_3
(C) SiH_4
D Cl_2O



8 The atomic numbers of elements X and Y are 1 and 7 respectively. Which of the following combinations concerning the compound they form is correct?

	Chemical formula	Type of bond
A	YX_2	covalent
(B)	YX_3	covalent
C	YX_2	ionic
D	YX_3	ionic

Element X is hydrogen while element Y is nitrogen. They combine to form a covalent compound ammonia, NH_3.

9 Consider the information given in the table below:

Element	Atomic number
w	6
x	9
y	10
z	12

Which of the following pairs of elements would react with each other most readily?

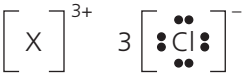
(A) w and x
B w and z
C x and y
D y and z

10 The atomic number of element X is 15. X reacts with chlorine to form a chloride. Which of the following can represent the electron diagram of the chloride? (Only electrons in the *outermost shells* are shown.)

A 

(B) 

C 

D 

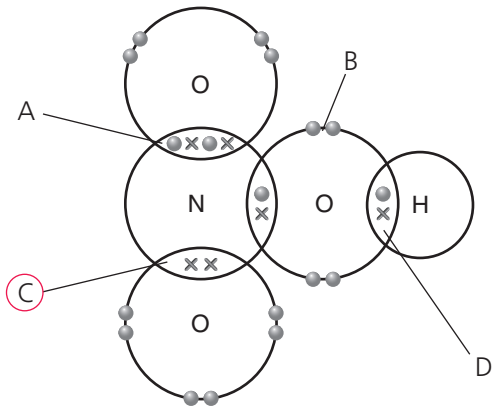
(HKCEE, Paper 2, 2011, 2)

9

Element	Atomic number	Name of element
w	6	carbon
x	9	fluorine
y	10	neon
z	12	magnesium

Elements w and x are non-metals. They combine to form a covalent compound.

11 The diagram below shows an electron diagram of nitric acid.



Identify which of the labelled sets of electrons represents a dative covalent bond.

(Edexcel Advanced Subsidiary GCE, Unit 2, May 2012, 2(a))

12 Which of the following compounds contains both ionic and covalent bonds?

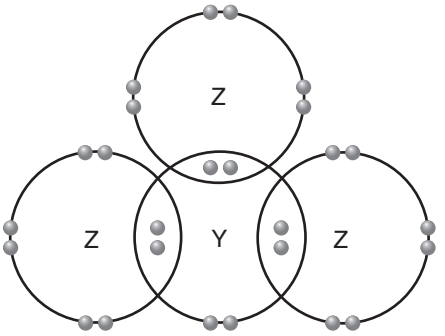
A $CaCl_2$
B BF_3
C NH_4Br The NH_4^+ ion contains a covalent bond. Ionic bond exists between NH_4^+ ion and Br^- ion.
D SO_2

13 In which of the following groups of substances there exists a difference in bonding type among the substances?

A Iodine, oxygen, nitrogen
B Chromium, mercury, aluminium
C Methane, ethyl ethanoate, sulphur dioxide
D Potassium chloride, hydrogen chloride, silver chloride

(HKCEE, Paper 2, 2008, 18)

14 Y and Z are two different elements. The electron diagram of the compound formed between Y and Z is shown below:



(Only electrons in the *outermost shells* are shown.)

Which of the following statements is / are correct?

(1) There are 5 electrons in the outermost shell of an atom of Y.
(2) There are 8 electrons in the outermost shell of an atom of Z.
(3) The atom of Y in the compound above has 4 lone pairs.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(2) There are 7 electrons in the outermost shell of an atom of Z.
(3) The atom of Y in the compound has 1 lone pair.

15 Which of the following statements concerning a water molecule is / are correct?

(1) The number of bonding electrons contributed by each hydrogen atom in the molecule is 2.
(2) The number of bonding electrons contributed by the oxygen atom in the molecule is 2.
(3) The total number of electrons in the molecule is 8.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2008, 19)

Part III Structured questions

16 Draw electron diagrams of the following molecules, showing electrons in the *outermost shells* only.
Refer to page T37.

a) Oxygen

b) Nitrogen

c) Carbon dioxide

d) Silicon tetrachloride

17 Chlorine is an element in Group VII of the periodic table.

a) i) The relative mass of every chlorine atom is given as a whole number and yet the element chlorine has a relative atomic mass of 35.5. Explain how this is possible.
Isotopes of chlorine exist.

ii) The atomic number of chlorine is 17.

Write the electronic arrangement of a chlorine atom.
2,8,7

b) Chlorine reacts with hydrogen to form hydrogen chloride, HCl.

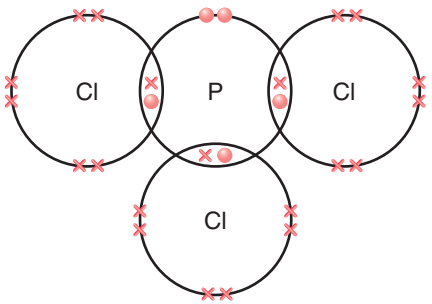
Draw an electron diagram of a hydrogen chloride molecule, showing *outermost shell* electrons only.

(Edexcel GCSE (Higher Tier), Chemistry, Unit C2, Mar. 2012, 4(a), (c))

18 Phosphorus trichloride, PCl_3, is a covalent compound.

a) Give a reason why the bond between phosphorus and chlorine is covalent.
The bond is formed between two non-metals.

b) Complete the diagram below to show the *outermost shell* arrangement of electrons in phosphorus trichloride.



(Edexcel GCE O Level, Paper 2, Jan. 2010, 4(b))

19 a) Explain how the atoms are held together by the covalent bond in a molecule of hydrogen.
Refer to page T38.

b) Draw electron diagrams of the following species, showing electrons in the *outermost shells* only.

i) Methane, CH_4

ii) Ethene, $CH_2{=}CH_2$

iii) Nitrogen, N_2

iv) Ammonium ion, NH_4^+

(Edexcel Advanced Subsidiary GCE, Unit 1, May 2012, 26(a)–(b))

20 a) The table below gives information about two isotopes of hydrogen, hydrogen-1 and hydrogen-2.

	Hydrogen-1	Hydrogen-2
Atomic number	1	1
Mass number	1	2

An atom of hydrogen-1 is represented as ^1_1H.

Show how an atom of hydrogen-2 is represented.
^2_1H

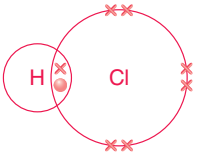
b) Calculate the relative molecular mass of water, H_2O.
2 x 1.0 + 16.0 = 18.0
(Relative atomic masses: H = 1.0, O = 16.0)

c) Molecules of heavy water contain two atoms of hydrogen-2 instead of two atoms of hydrogen-1.

Explain why a molecule of heavy water has more mass than a normal water molecule. You should refer to the particles in the nucleus of the two different hydrogen atoms in your answer.

(AQA GCSE (Higher Tier), Chemistry, Unit 2, Jan. 2010, 3(a)–(b)(i), (c))
An atom of hydrogen-2 has a neutron while an atom of hydrogen-1 has none.

17 b)



21 Chemists have developed models for bonding and structure which are used to explain different properties. Refer to page T39.

a) Ammonia, NH_3, is a covalent compound.

i) Explain what is meant by a 'covalent bond'.

ii) Draw an electron diagram to show the bonding in NH_3, showing electrons in the *outermost shells* only.

b) Ammonia reacts with hydrogen chloride, HCl, to form ammonium chloride, NH_4Cl.

NH_4Cl is an ionic compound containing NH_4^+ and Cl^- ions.

i) Write down the electronic arrangement of the Cl^- ion.

ii) Draw an electron diagram to show the bonding in NH_4^+, showing electrons in the *outermost shells* only.

(OCR Advanced Subsidiary GCE, Chem. A, F321, Jan. 2009, 2(a)(i)–(ii), (b)(i)–(ii))

22 When H^+ ions react with H_2O molecules, H_3O^+ ions are formed.

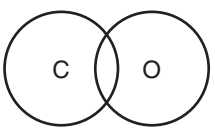
Name the type of bond formed when H^+ ions react with H_2O molecules. Explain how this type of bond is formed in the H_3O^+ ion. Type of bond: Dative covalent bond

(AQA Advanced Subsidiary GCE, Unit 1, Jan. 2010, 3(e))
Explanation: The lone pair on the oxygen atom provides both the electrons for bonding with the H^+ ion.

23 Carbon forms an oxide called carbon monoxide (CO). A dative covalent bond exists between the two atoms.

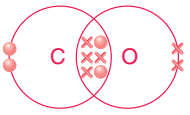
a) Define the term 'dative covalent bond'.

b) Complete the diagram below for a CO molecule using dots or crosses to represent electrons. Just show all the *outermost shell* electrons.



23 a) A dative covalent bond is a covalent bond in which the bond pair electrons are provided by the same atom.

b)



24 Predict the type of compound (ionic / covalent) formed when each of the following pairs of elements combine. Draw electron diagrams of the compounds formed, showing electrons in the *outermost shells* only. Refer to page T39.

a) Phosphorus and hydrogen

b) Aluminium and fluorine

c) Silicon and chlorine

d) Carbon and sulphur

25 The non-metallic element selenium (Se) has six outermost shell electrons in its atom.

a) To which group in the periodic table does selenium belong?
Group VI

b) Hydrogen selenide is a compound made up of hydrogen and selenium.

i) Predict the type of bonding present in hydrogen selenide.
Covalent bond

ii) Draw an electron diagram of hydrogen selenide, showing electrons in the *outermost shells* only.

26 The table below gives the melting points of Group VII elements. Refer to page T40.

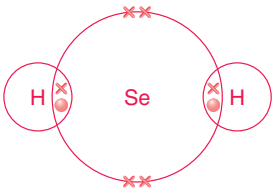
Element	Atomic number	Melting point (°C)
Fluorine	9	–220
Chlorine	17	–101
Bromine	35	–7
Iodine	53	113
Astatine	85	302

a) What is the name given to Group VII elements of the periodic table?

b) i) Explain why elements in Group VII exist as diatomic molecules.

ii) Name TWO other elements apart from those in Group VII which are diatomic.

25 b) ii)



c) Draw an electron diagram of the compound formed from fluorine and chlorine, showing electrons in the *outermost shells* only.

d) Draw an electron diagram of the compound formed from chlorine and oxygen, showing electrons in the *outermost shells* only.

27 The following table lists the atomic numbers of four elements.

Element	Atomic number
W	7
X	9
Y	10
Z	20

a) Suggest why element Y does NOT form compounds.

b) P is a compound formed from elements W and X.

i) Predict the type of compound (ionic or covalent) formed.

ii) Draw an electron diagram of P, showing electrons in the *outermost shells* only.

iii) State the chemical formula of P.

c) Q is a compound formed from elements X and Z.

i) Predict the type of compound (ionic or covalent) formed.

ii) Draw an electron diagram of Q, showing electrons in the *outermost shells* only.

iii) State the chemical formula of Q.

28 Calamine lotion is used to treat itching. The main ingredients are two metal oxides.

a) One of the metal oxides has a formula mass of 81.0, and the chemical formula of this metal oxide is MO. (M is NOT the correct symbol for the metal.)

The relative atomic mass of oxygen is 16.0.

i) Calculate the relative atomic mass of metal M.

ii) Use your answer to part (a)(i) and the periodic table to name metal M.

b) The other metal oxide is iron(III) oxide, which contains iron(III) ions (Fe^{3+}) and oxide ions (O^{2-}).

Explain, in terms of electrons, how an iron atom (Fe) can change into an iron(III) ion (Fe^{3+}).

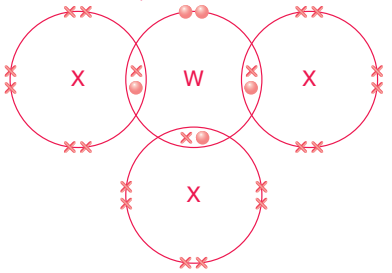
(AQA GCSE (Higher Tier), Chemistry, Unit C2, Jan. 2011, 2(a)–(b)(i))

Unit Exercise

27 a) An atom of Y has 8 electrons in its outermost shell which is extremely stable.

b) i) Covalent compound

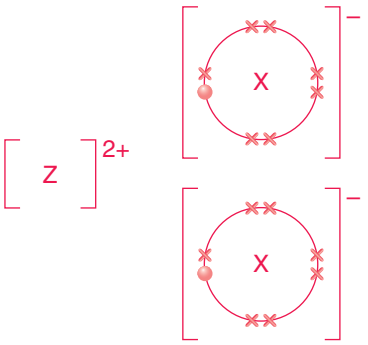
ii)



iii) WX_3

c) i) Ionic compound

ii)



iii) ZX_2

28 a) i) Formula mass of MO = 81.0
= relative atomic mass of metal M + 16.0
relative atomic mass of metal M = 65.0

ii) The name of metal M is zinc.

b) When an iron atom loses three electrons, an iron(III) ion forms.